

Homework 1

MATH 346/377 - Winter 2025

Due on Sunday, January 26, 11:59pm on Crowdmark

Total of points = 80 points

A complex number $\alpha \in \mathbb{C}$ is an *algebraic number* if there exists a polynomial $f \in \mathbb{Q}[x]$ such that $f(\alpha) = 0$. A *minimal polynomial* of α is a polynomial in $\mathbb{Q}[x]$ of minimal degree that has α as a root. Such a polynomial is not unique (you can always multiply by an element in $\mathbb{Q}^\times$ to get another minimal polynomial). In the textbook and in the first lecture notes, to get a unique minimal polynomial we required that its coefficients are coprime integers and that the leading coefficient is positive. Sometimes, it is also convenient to replace¹ this condition by the condition that the minimal polynomial is *monic* (in that case the coefficients are a priori only rational numbers, not integers). An algebraic number $\alpha \in \mathbb{C}$ is an algebraic integer if it has a monic minimal polynomial with integer coefficients.

For the following exercise you might need the following Lemma, also known as Gauss' Lemma. It says that if an integral polynomial can be factored over $\mathbb{Q}[x]$, it can be factored over $\mathbb{Z}[x]$.

Lemma 0.1. *Let $f(x) \in \mathbb{Z}[x]$ be a polynomial with integer coefficients and suppose that we can factor $f(x) = A(x)B(x)$ with $A(x), B(x) \in \mathbb{Q}[x]$ two polynomials. We can find $r, s \in \mathbb{Q}$ such that $a(x) = rA(x)$ and $b(x) = sB(x)$ are in $\mathbb{Z}[x]$ and $f(x) = a(x)b(x)$.*

1. (14 points) Show the following.
 - (a) (5 points) Let $m(x) \in \mathbb{Q}[x]$ be a minimal polynomial of an algebraic number $\alpha \in \mathbb{C}$. If $f(x) \in \mathbb{Q}[x]$ has α as a root, then $m(x)$ divides $f(x)$.
 - (b) (3 points) Deduce that an algebraic number α has a unique monic minimal polynomial.
 - (c) (6 points) An algebraic number $\alpha \in \mathbb{C}$ is an algebraic integer if and only if α is the root of *some* monic polynomial with integer coefficients. (So not necessarily the minimal one).

¹Given a minimal polynomial that has coprime integer coefficients but is not monic, one obtains a monic minimal polynomial with rational coefficients by dividing by the leading coefficient. Conversely, given a monic minimal polynomial with rational coefficients, we get a minimal polynomial with coprime integer coefficients by clearing the denominators. For example $2x^2 - 1$ and $x^2 - 1/2$ are both minimal polynomials of $1/\sqrt{2}$. The first one has coprime integer coefficients, the second one has rational coefficients but is monic.

2. (13 points) The goal of this problem is to show that the set of algebraic integers form a ring. We say that a ring R is finitely generated over $\mathbb{Z}$ if there exist elements $r_1, \dots, r_k \in R$ such that every element $r \in R$ can be written

$$r = n_1 r_1 + \dots + n_k r_k \quad (2)$$

for some integers $n_1, \dots, n_k$ in $\mathbb{Z}$.

- (a) (0 points) If $R \subset \mathbb{C}$ is finitely generated and $\alpha \in R$, then α is an algebraic integer. *This part is trickier and you can use it without proof in the rest of the exercise. This part will not be marked but if you want to prove it, the idea is to represent α by a matrix and to use Cayley-Hamilton theorem.*
- (b) (6 points) Show that if $\alpha \in \mathbb{C}$ is an algebraic integer, then

$$\mathbb{Z}[\alpha] = \{f(\alpha) \mid f \in \mathbb{Z}[x]\} = \left\{ \sum_{i=0}^d a_i \alpha^i \mid a_i \in \mathbb{Z} \right\} \quad (3)$$

is finitely generated.

- (c) (4 points) Use the same idea to show that $\mathbb{Z}[\alpha, \beta] = (\mathbb{Z}[\alpha])[\beta]$ is finitely generated if $\alpha, \beta \in \mathbb{C}$ are algebraic integers.
- (d) (3 points) Deduce that $\alpha + \beta$ and $\alpha\beta$ are algebraic integers.

Pari GP is open-source computer algebra system designed for fast computations in number theory. You can run it on your browser at the following link <https://pari.math.u-bordeaux.fr/gpasm.html>.

3. (19 points) The command `algdep(x,d)` takes as input a number x (computed to some accuracy), an integer d and tries to find a polynomial of degree d whose coefficients are as small as possible and for which x is a root.
- (a) (2 points) The accuracy is determined by the command `\p`. Display π with an accuracy of 5 digits: first type `\p5` followed by `Pi`. Vary the precision to get a sense of how it works. Set the precision to 5 and use the command `algdep(Pi,6)`. What do you observe and what does it suggest? Change the precision to 500 digits and run the command again.
- The goal of this question is to show the limitations of numerical computations. If the accuracy is very high compared to the degree, the command `algdep(x,d)` can give strong evidence that a number is algebraic, but it does not prove that it actually is.*
- (b) (4 points) From now on set the accuracy to 500 digits. By Exercise 2 the real number

$$x = \sqrt{2} + \sqrt{3} \quad (7)$$

is an algebraic integer. Run the command `algdep(x,d)` and increase the input d . Find a possible minimal polynomial of x , and verify by hand that x is indeed a root.

Remark: the command for $\sqrt{2}$ is `sqrt(2)`.

- (c) (4 points) Show that if $\alpha \in \mathbb{C}$ is a nonzero algebraic number, then $\frac{1}{\alpha}$ is an algebraic number. Use Pari GP to find a minimal polynomial of

$$x = \frac{1}{\sqrt{2} + \sqrt{3}}. \quad (8)$$

What do you observe? Write down an example of an algebraic integer α such that $1/\alpha$ is *not* an algebraic integer.

- (d) (4 points) Show that if a real number α is an algebraic integer, then $\sqrt{\alpha}$ is also an algebraic integer. Deduce from Exercise 2 that

$$x = \sqrt{\sqrt{3} + \sqrt{2}} + \sqrt{\sqrt{3} - \sqrt{2}} + 2i. \quad (9)$$

is an algebraic integer and find a minimal polynomial of x using Pari GP (you do not need to verify your result by hand).

Remark: the command for $a + ib$ is `a+b*I`.

- (e) (5 points) Let $t \in \mathbb{Q}$ be a rational number. Show that $2\cos(\pi t)$ is an algebraic integer. Use Pari GP to find a possible minimal polynomial of $\cos(3\pi/5)$.

Remark: the command is `cos(Pi*3/5)`. You don't have to verify that $\cos(3\pi/5)$ is indeed a root but you can try!

4. (6 points) Find the gcd of 939 and 753 and find $u, v \in \mathbb{Z}$ such that

$$939u + 753v = \gcd(939, 753). \quad (14)$$

Verify your solution.

5. (10 points) Find a gcd d of $2 + 5i$ and $7 - i$ and find $\alpha, \beta \in \mathbb{Z}[i]$ such that

$$\alpha(2 + 5i) + \beta(7 - i) = d. \quad (25)$$

Verify your solution (using Pari GP for example).

6. (18 points) The ring of integers of the field $\mathbb{Q}(\sqrt{-3})$ is the ring of Eisenstein integers

$$\mathbb{Z}[\omega] = \{a + b\omega \mid a, b \in \mathbb{Z}\} \quad (36)$$

where $\omega = \frac{1+\sqrt{-3}}{2}$.

- (a) (4 points) What is the monic minimal polynomial of an element $a + b\omega$?
- (b) (4 points) What are the units of $\mathbb{Z}[\omega]$?
- (c) (10 points) Show that $\mathbb{Z}[\omega]$ is a Euclidean domain.